

ActInf GuestStream 110.1 ~ Triple equivalence for the emergence of biology

GuestStream | Triple equivalence for the emergence of biological intelligence (Isomura) | 1:13:08

Hello, welcome. It is July 3rd or 4th, depending on where you are, and we are in Active Inference Guest Stream 110.1 with Takuya Isomura discussing the triple equivalence for the emergence of biological intelligence. Previously, we had some discussions in the Livestream 51 series when we visited Area 51 together, and we talked a lot about a double equivalence, and now the bar has been raised, the increment has been incremented, and this is a really exciting area that I'm sure we'll have a lot to discuss and learn from. So thank you again for joining, and looking forward to the presentation.

Great. Thank you, Daniel, for the invitation. Thank you for inviting me for this guest talk of Active Inference Institute. I'm very happy to share my recent paper on the triple equivalence.

So let's get started. So I'm interested in the theory of intelligence in general. So maybe the baby has no such high intelligence, but through learning, it obtained the intelligence. This can be modeled or understood in terms of the self-organization of neural network. So newborn neural network may not have enough information regarding the external world, but through some optimization, which is typically expressed in the world. So the energy minimization. It achieves some

It achieves some

It achieves some

Obtaining

Obtaining

of some

Good structure to

Recapitulate the external world, which is the basis for prediction and making insight or making creativity. So we would like to model those emergence of intelligent function within the neural network. Through the NMV energy minimization. So far, there are.

Energy minimization. So far there are. To my opinion, there are some

Three major pillars in the theory of intelligence.

The theory of intelligence. So far, there are three major pillars in the theory of intelligence.

So intelligence. So intelligence. So intelligence. So intelligence.

So intelligence. Occurs in the brain. So we

Model the brain using the network of neural

this point represents a solution of Bayesian theorem.

So instead of directly computing equation,

by just minimizing energy,

we can get the best guess about the external world.

And this process is expressed like this.

So this is the idea behind the theory called the free energy principle.

So free energy principle, as you may know,

a theory of the brain proposed by Carl Fristen,

which states that perception, learning,

and action of all biological organs are determined to minimize variational free energy.

And by doing so,

they perform variational Bayesian inference for external world.

This animation is the one I often use to explain the free energy principle framework.

So here there is an animal,

which is our agent that infers external world.

An external world comprises observable information,

and unobservable information,

which is a hidden state.

And the task for this agent is to infer the hidden state based only on observable information.

And by constructing some prior belief in the,

sorry,

by constructing the posterior belief in the brain,

it can guess what is the current hidden state.

And this can be done by minimizing the free energy.

So idea is that if we change the posterior belief S to minimize this landscape,

then we get the optimal solution,

which is called posterior belief or posterior expectation.

And this is an equation of the variational free energy.

It comprises the prediction error term and entropy or complexity term.

So to achieve this sort of influence,

we also need to update the model itself.

So this is done by update of the parameter θ .

So in the external world,

there is some mapping from hidden state to observation.

So we imitate this mapping using the generative model.

So by tuning this θ ,

our generative model is optimized to recapitulate the external world.

And then we get the best guess about the hidden state.

This is a mechanism underlying the perception.

And the unique point of the free energy principle is that it can be applied also to the optimization of action.

So here we consider some,

we consider that there is some preference prior for this agent.

And to achieve these sensory inputs,

this agent can select an action from a given set.

Here there are three options.

And this optimization can be done by minimizing the free energy in the external, free energy in the future,

which is called expected free energy.

So expected free energy is a function of action.

So by selecting the action that achieves a minimum free energy,

minimum expected free energy,

it can maximize the probability to obtain the preferred input in the future.

So this is the idea,

a so-called active inference.

So like this,

the free energy principle is a normative theory

that can express both perceptual optimization and action optimization.

But for neuroscientists like us,

the most interesting point is the,

what is the neuronal implementation underlying those inference?

So we would like to show

actual neuronal implementation of this sort of active inference.

So to address this issue,

we recently proposed equivalence,

double equivalence between neuron dynamics and Bayesian inference.

So we start from evaluating the neural activity equation.

So neural activity equation is,

have long history.

So it's have a clear biological background.

So we consider that this is a plausible one.

And if this equation is derived as a gradient descent of some energy function,

we can obtain the original equation by just taking the integral of this activity equation

and get energy, which is called Helmholtz energy.

On the other hand,

for Bayesian inference,

once we define the model,
generative model about the external world,
we can derive the free energy function.
This is a variational free energy.
And its minimization indicates the achievement of Bayesian inference.
We show a mathematical equivalence between those two energy functions,
which means that for any neural network in this class,
there is an interpretation in terms of Bayesian inference.
So we can, in general,
see neural activity or neural network dynamics,
including synaptic plasticity,
as an inference about the external states.
So this is a more mathematical definition of our equivalence.
So we characterize neural activity equation using some rate coding model,
which is derived from a realistic spiking model through some approximation.
So its form is like this.
It is characterized by inverse sigmoid liquid factor,
and all indicate the sensory input from input layer.
 W is a synaptic weight.
 H is a firing threshold.
And X here is a vector of neural activity,
which is a rate coding neurons.
And its dynamics is given like this.
And if this is given,
then we can consider the original energy function
by taking the integral of this activity equation with respect to X .
And we get an explicit form of Helmholtz energy like this.
Although it is a little bit complicated,
it's just the integral of the original equation.
So don't worry about that.
And those definitions of $\bar{\cdot}$ is here,
and the $\hat{\cdot}$ indicates the sigmoid function of W .
And ϕ is a threshold factor defined like this.
And interestingly, although this energy is purely derived from activity equation,
the same energy provides the plasticity equation.
If we take the derivative of this A with respect to W ,
then we get a Hebbian plasticity equation.
So this is non-trivial, right?

This is purely derived from this activity equation,
but it also provides plasticity.

Moreover, for a Bayesian inference,
once we define the generative model,
we get the variational free energy.

So this is a variational free energy under the assumption that
the generative model have the shape of the partially observable
Markov decision process or HOMDP.

So if the external world is expressed as a discrete state space,
then we get this kind of variational free energy.

And its minimization provides the solution of Bayesian inference,
which is a posterior brief about state.

And we also obtain the posterior brief about the model parameter.

So which correspond to learning.

And what we say is that there is a formal equivalence between those two energy function
in the sense that each component have the one-to-one mapping between those two.

So S here indicates the posterior expectation about each state taking 0 or 1.

So it is expressed as the block vector, just like the block vector of X and $\bar{X}$.

And this logarithm corresponds to this logarithm.

And this A actually has the block matrix structure, which is exactly corresponding to this block matrix.

And this operation in a product corresponding to this product of matrix and vector.

And this $\log D$, which is the state prior, is formally corresponding to ϕ here.

So as a result, we can say that for any neural activity and plasticity in this class that minimize the
common Helm-Holtz energy, we can be, we can lead those dynamics in terms of the Bayesian inference of
external states.

Okay. And moreover, this framework can be extended to the empirical application.

So to apply the theoretical framework into the empirical data, we first record neuronal activity from real brain.

And we assign those data into our canonical neural network, which is our network model.

And by taking the integral, we can get the underlying Helm-Holtz energy, which characterize the neural
dynamics and plasticity.

Moreover, we can estimate the parameter from data to characterize for, characterize this energy function for
specific for individual.

And through our established equivalence, we can identify the generative model and the corresponding
operational

fluid energy from this A .

And this works as a synthetic agent that perform active inference.

So it's deliberately produce a learning algorithm or plasticity algorithm that update sine-artic weight to adapt
to the environment.

And it's time integral provides a prediction about the learning process.

So we call this framework as a reverse engineering in the sense that from neural activity data, we can get the generative model, which is a blueprint for the individual animals.

So by doing that, we can get the synthetic agent that imitates the original animal, animal activity and behavior.

So now we can make the examination of the free energy principle.

So our idea is that, so we can do this based only on neural activity in the initial learning stage.

So we record neural activity before training and characterize generative model and make prediction of the subsequent learning process based on the synthetic agents that perform active inference.

So if its prediction is consistent with neural activity data recorded after training, then it indicates the possibility of the theory underlying this synthetic agent.

So if this is the case, then we can validate the, we can validate the predictive validity of the free energy principle and active inference.

So this is the idea.

So we apply this or we apply this to in vitro neural network, cultured on a dish.

So this is a neuron taken from rat embryos.

And we culture those neurons on a dish that have 64 electrodes at floor.

And we stimulate neurons using designed sensory signals, which is the electrical pulse.

So to stimulate neurons, we design those stimuli, stimulation pattern using two hidden states.

So those are hidden states or hidden sources that is invisible for in vitro neural networks.

So they can receive signal, directly receive signals only from sensory stimuli here.

So by receiving those mixed sensory signals, those neurons are required to infer the hidden state.

So if it can infer the hidden state, then it supports the prediction of free energy principle.

So we test that.

So we found that actually some neuron response preferentially to a specific source.

So the prediction of free energy principle is correct in this setting.

And we also statistically estimate the effective connectivity between those neurons and the plot those trajectory.

So synaptic weights change during the training.

So we plot that trajectory onto the theoretically derived free energy landscape.

So this is a free energy characterized only by the initial data of this in vitro learning experiment.

And we now plot the trajectory onto this.

So this is the empirically estimated synaptic weight trajectory, which started from the high point of the landscape and it reduced to the reduce the free energy to reach the bottom, which is a consistent with the expectation of the free energy principle that

synaptic plasticity occur to minimize the free energy.

So yeah, it's consistent.

And if this is the case, we can make synthetic agents that perform free energy minimization.

So this is a result of the simulator that just exploit the gradient of this theoretically computed free energy landscape.

So it is just a simulation without using data.

Nevertheless, its result is highly consistent with actual observation, meaning that the theory has the prediction ability in this setting.

So it shows the validity of the free energy principle.

So, and then we extend our idea to action optimization, active inference.

To optimize action, we need to consider the modulation of learning and plasticity because our action is changed based on the given reward or punishment.

So, uh, uh, classically, this sort of optimization of action, uh, is associated with the learning and plasticity modulation by a neural modulator, various neural modulator.

One, uh, famous one is a dopaminergic modulation of learning and type plasticity.

So it changed the time window of the STDP.

So STDP here means the, uh, spike timing dependent plasticity, which is a sort of, uh, learning and plasticity.

So it changed, dopamine changed drastically the window of STDP like this.

And we incorporate those effects into the, uh, into the plasticity equation.

And we also consider two layer recurrent neural network, uh, with the output layer to model, uh, the action of animals.

So it is, uh, basically same, uh, model as the previous one, uh, we call this, uh, canonical neural network.

And, uh, one, uh, important, uh, characteristic of this canonical neural network is that all of those equations, both neural activity and plasticity can be derived from a single energy function, which means that there is a corresponding Bayesian inference, uh, characterized by variational free energy.

So we, again, uh, get the one-to-one correspondence between neural dynamics and Bayesian inference.

Interestingly, this sort of neural network can perform, uh, maze task.

So this is a typical delayed reward task, but by, uh, inferring the best action that obtain the, uh, maximum reward in the future.

So which is, uh, in other words, uh, minimum risk, uh, is associated with the future outcome.

Uh, by doing so, this network can, uh, achieve the best action, uh, to, uh, realize, uh, this maze task.

So, so far we show, uh, double equivalence between neural network dynamics and Bayesian inference.

So in general, uh, neural network, canonical neural network that minimize some energy function by both activity and plasticity, it inevitably perform, uh, Bayesian inference, active inference, uh, in some sense.

So, uh, uh, free energy principle is such a generic, uh, general framework to characterize, uh, this sort of biological approach for neural network.

The next step is, uh, incorporation of the algorithm computation, because so far I discussed about, uh, optimization of perception and action, which is not, uh, completely generic, uh, intelligence form.

So it is, uh, limited, uh, only for optimizing, uh, behavior to obtain, uh, to, to minimize the future risk.

But question is whether this framework can be applied to the emergence of a generic intelligence, generic algorithm, that's, uh, my interest.

So to address this issue, we propose the triple equivalence.

We show the triple equivalence between neural network dynamics,

and variational Bayesian inference framework, and the Turing machine.

So here, uh, we have two Turing machines.

One is the internal Turing machine, and one is the external Turing machine.

So instead of characterizing environment in terms of the generative model,

uh, we characterize the, uh, the environment in terms of external Turing machine.

But the point here is that such a Turing machine can also be implemented in terms of the probability, uh, distribution.

So, which is actually expressed in terms of generative model, uh, and more well,

because Turing machine typically deal with the, uh, deal with the discrete state space.

So this is expressed in the form of POMDP.

So we simply consider the POMDP with some, uh, external tape.

So this is, uh, what, uh, I actually did.

So to infer such environment, we have the, uh, Turing machine, which is basically have the same shape as the external Turing machine.

And we show that this type of Turing machine can be implemented by, uh, canonical neural network.

So idea, uh, is that so we first, uh, characterize the canonical neural network in terms of the energy function like this.

And we also characterize differentiable Turing machine, uh, in terms of the, in terms of the, uh, energy minimization.

So if we get, if we write, uh, Turing machine in some differential equation by taking the integral, uh, we can again obtain the energy function.

So, and through this process, we confirm that actually the energy function derived from Turing machine is equivalent to that

for canonical neural network.

So this is, this happened because, uh, output layer of canonical neural network can be seen as a tape, uh, for a Turing machine.

And this operation, reading out and writing into tape can be implemented, uh, in a biologically plausible manner through, uh, mental action that, uh, affects

some, uh, some, uh, change for the internal state.

And the writing can be implemented by rapid plasticity of output layer, uh, symmetric weight.

So through those process, uh, we can, uh, implement a function of Turing machine, which is the, uh, reading and writing and state update.

And header position movement is also implemented by the neural activity, uh, sparse neural activity that, uh, expressed, um, address of the information.

And, uh, once we get, uh, energy function, which is common for, uh, all, uh, component in the system, then we can say that there is a corresponding variation of free energy.

So, uh, we found that actually a disorder of Turing machine, uh, formally correspond to, uh, kind of, uh, POMDP generative model.

And, uh, you know, again, uh, this mapping from, uh, output to, uh, state posterior corresponded to, uh, read

out or, uh, information, read out of information.

And, uh, uh, if we rapidly change the C matrix, which is a policy matrix that determined our action, then, uh, this, uh, act as memory writing.

So, rapid change of, uh, policy, uh, can be seen as, uh, writing of information.

So, in short, what we show is that we characterize neural network in terms of the energy minimization.

So, it involves both neural activity and plasticity.

Interestingly, this typical neural network dynamics can be read as both mapping, uh, of Turing machine and Bayes-Shera.

In the sense that, so this is, uh, energy minimizing activity.

And, uh, if energy is minimized with respect to internal state of neural network, then this solution, uh, can be seen as the solution of Bayesian inference, which is a Bayes-Shera.

And also, in the, simultaneously, it also represents, uh, mapping, uh, dismapping.

Which means that, uh, this network is not limited to infer the, uh, POMDP, typical POMDP in the external world.

But also, it can infer the Turing machine in the external world and imitate that function into, uh, inside the network, uh, dynamics.

So, it can, uh, imitate, uh, mimic the external algorithm using neural network dynamics.

So, that's the idea.

So, uh, based on this generic framework, uh, we perform some demonstration.

So, this is a, uh, demonstration using other.

So, other means a machine that adds count up the number.

So, here we provide the input as the, uh, MNIST hand digit.

So, actually, this is only binary signal.

So, each, uh, becomes either zero or one.

And this, uh, 16, uh, 16, uh, binary, uh, number represents, uh, inputs.

And the external Turing machine count up those numbers and memorize that sum, uh, into the tape.

So, this is what the environment works.

Given this system, the internal canonical neural network, which perform active inference, can perform the imitation of these dynamics by receiving this hand digit.

It infer whether this is zero or one through, uh, this, uh, layer.

And the internal computation make addition of those, uh, signal into the, uh, previous summation of the, uh, which is, uh, stored in the memory.

And the memory is actually implemented by output layer neural, uh, output layer synaptic weights.

So, here, uh, by, by, uh, given the internal state, the middle layer neural activity,

it, uh, sends some signal to output layer, which corresponds to address of the, uh, information.

And given this, uh, output layer activity generated action, uh, this weight corresponds to the memory, uh, in the tape.

And it takes, uh, either zero or one.

So, by, by sending your signal header information here, uh, it receives, uh, the memory information.

And it feeds back to the internal state.

So, this is, uh, how this network works as an other.

And, uh, writing of new information into the output layer is done by heavy and plastic, uh, modulated by, uh, risk factor.

So, this, uh, plasticity occur in a rapid manner.

So, that, uh, by receiving new risk, it, uh, drastically change the synaptic weight, either from zero to one or one to zero, which is sufficient to implement, uh, the writing function of the Turing machine.

So, this sort of neural network can implement, uh, Turing machine in a biological plausible manner.

And, uh, this is a result of inference.

So, by receiving hand rigid, hand, hand digit, which is, uh, noisy information, it can correctly infer the hidden state, uh, like this.

And, uh, it successfully store the information in the output layer, uh, which represents the summation of the number.

So, it counts up the input number and the store that here.

And through learning, the accuracy of inferring, uh, hidden state and memorize, uh, information, uh, summation, some, some, some, some number is improved.

So, before training, uh, the estimated number is largely different from the true number.

But, uh, in these four sessions, uh, which comprises, uh, 2048 steps, it achieved, uh, good inference.

And, uh, uh, this is a quantitative, uh, evaluation of prediction error.

Uh, it decreased, uh, uh, uh, time increase, uh, uh, uh, time increase.

And also this network optimize its parameter by minimizing free energy.

And, uh, uh, the parameter estimation error decrease with time like this.

Okay.

Um, this is, uh, this is a demonstration of our algorithmic implementation and inference of external algorithm using canonical neural network.

However, there is some limitation in this framework.

So, here we assume, uh, predefined learning, uh, sorry, predefined writing rule, which is implemented by risk function.

So, if we can design a risk function or which is corresponding to expect free energy a priori, okay, then we can train the neural network in an appropriate manner.

But question is whether, uh, those predefined function can also be optimized through a biological plausible, uh, process.

That's our next question.

To address that, we consider, uh, extension of our framework to the evolution of time scale.

So, evolution occurs through the update of gene and we associate that with the update of model.

So, we now extend the framework to the Bayesian model selection framework.

So, uh, by updating the model, uh, we get a better generative model.

And based on that model, we update parameter through the synaptic plasticity and then infer the hidden state and action, uh, through neural activity.

And, uh, its result is evaluated and then, uh, it's, uh, iterated, so on, so.

So, we model this framework using extended Helm-Holtz energy, which is a function for a species, not the individual.

So, we now define, uh, the distribution of agent like this.

So, this represents, uh, distribution, uh, or over observation sequence and gene distribution.

So, which characterize, uh, what kind of agent exists in, uh, environment.

Okay.

And how, what kind of information received, uh, is received, uh, by those agents.

So, we now consider the populations.

So, this is defined, uh, by adding, uh, some component to the individual Helm-Holtz energy, uh, defined previously.

So, this is, uh, um, Helm-Holtz energy, uh, we discussed.

And, uh, we add here to some factor.

For example, this, uh, represents the reproduction rate, which is the evolutionary fitness function.

And this is, uh, uh, parent gene distribution, which correspond to prior distribution of the gene here.

And this is, uh, uh, uh, entropy, uh, but it is, uh, representing, it represents, um, posterior belief, uh, about, uh, the observation, the gene distribution.

So, it is analogous to, uh, conventional variational free energy, but now we, uh, also optimize the model.

So, interestingly, by, uh, taking the derivative of this, or taking the variation of this, uh, functional, we get a solution like this.

So, this solution, of course, it has a meaning in terms of Bayesian influence, but it has another meaning.

So, interestingly, this, uh, solution, uh, posterior belief represents, uh, uh, is given as the product of, uh, evolutionary fitness function and marginal likelihood and prior gene distribution.

And this marginal likelihood can be seen as the observation, uh, conditioned by the gene.

So, this result actually is equivalent to, uh, the solution of, uh, natural selection, simple natural selection.

So, now we found that, uh, by defining Helm-Holz energy, which is, uh, equivalent to variational free energy, uh, uh, and minimizing that we get the solution of, uh, natural selection in a, uh, natural manner.

So, uh, this indicates that the solution, you know, uh, Bayesian model selection involves, uh, conventional natural selection.

And, uh, this can be also understand by this, uh, inequality.

So, here it indicates that, uh, population level Helm-Holz energy or, uh, variational free energy upper bounds, uh, this factor.

So, this is a negative logarithm of, uh, partition function z .

And interestingly, under this framework, this partition function, uh, correspond to the total offspring number.

So, uh, uh, minimization of free energy, uh, provides the maximization of total offspring number.

So, if, uh, this sort of, uh, agent want to maximize their offspring, then it's, uh, naturally

So in summary, we showed that dynamics of canonical neural network that minimize shared Helmholtz energy can be cast as minimizing variational free energy. So it's basically perform active

inference in general. And canonical neural network with mental action and rapid, modulated plasticity can implement Turing machine in a biologically plausible manner and perform variational Bayesian inference of external algorithm, which is expressed by a Turing machine. And if we extend Helmholtz energy minimization to the evolution of scale, then spacey level Helmholtz energy minimization involves natural selection. So that that process can be naturally derived from minimizing energy, providing a possible explanation for the emergence of generic biological intelligence. So finally, I just briefly mentioned that those programs, I mean, Matlab or Python codes are all available at my GitHub page. So if you are interested in please check that. Thank you.

Thank you. Awesome. A lot to cover there. For people watching live, they can write a question in the chat. I wrote down a few as well. So I'll just sort of start in no particular order.

Are we seeking to find pairwise transformational edges and then maintain overall network equality through transitive properties?

Or do you see these different equivalences as arising from a more general form or notation?

Thank you for your question. Can I see that comment?

Yes. Yep. Give me one second.

Yes. Yes. First part is a little bit complicated to me. So pairwise...

When you did the work in the double equivalence and now with the triple equivalence, the double equivalence, do you seek to find pairwise strong relationships and keep everything equal through the transitive property?

Or do you think that there could be a common representation that has more of like a hub and spoke topology with a common underlying format that then is connected to each of these domains?

So I'm not sure if I'm correctly understand the question, but unlike double equivalence, which is correspondence.

So in double equivalence, all neural network property, as long as it minimizes shared energy function, is always Bayesian inference. So there is a set of Bayesian inference. It is very general.

And there is a subpopulation, which is biological plausible Bayesian inference.

So all of those subpopulation is involved in the generic Bayesian inference.

So if we design neural network that by considering the energy minimization, then it always is involved in generic Bayesian inference framework. On the other hand, Turing machine or in particular, universal Turing machine is special one. So not all Turing machine works as universal Turing machine.

So we may design some random network or random machine, and we can say this is a Turing machine.

It's not, I mean, it's in some sense correct. In some sense, it's correct, but it's not useful so far.

I mean, universal computation, Turing completeness means the computation computing machine have sufficient

ability to deal with algorithm computation. So yeah, that is something outside.

So what we are now thinking is that there is a group that achieved the Bayesian inference, and there is another group that performed the Turing complete computation, and there is some overlap here. So if we focus on that, that would be interesting to consider the emergence of high level intelligence.

I don't know.

Okay. Awesome. All right. I'm gonna paste into the chat so you can see it.

I just wanted to read this quotation from a...

Right.

I never checked that.

Okay. This is from a 2017 article by Andre Carpathi called Software 2.0.

And so he wrote, I sometimes see people refer to neural networks as just another tool in your machine learning toolbox. They have some pros and cons. They work here and there, and sometimes you can use them to win Kaggle competitions. Unfortunately, this interpretation completely misses the forest for the trees. Neural networks are not just another classifier. They represent the beginning of a fundamental shift in how we develop software. They are software 2.0. And then a later section is,

it turns out a large portion of real world problems have the property it is significantly easier to collect data or more generally identify desirable behavior than to explicitly write the program.

Because of this and many other benefits of Software 2.0 programs that I will go into below, we are witnessing a massive transition across the industry where a lot of 1.0 code is being ported into 2.0 code. Software 1.0 is eating the world and now AI software 2.0 is eating software.

So how do you, how do you think the kinds of analytical and empirical results that you are yielding might play into how we think about software engineering and designing different kinds of intelligent systems? That's, that's a very interesting question. Yeah.

Right. So, so, so, thought of neural network, we, I'm, I'm working on is, yeah, largely different from the state of the

project, deep learning machines. So we are more focusing on how the real brain, how real brain works when it's doing some computation or making decision. And yeah, although the last gap between the neural network model, the neural network model in computational neuroscience, which is too simple, and the state of the state of the state of the AIs. So we think that if we want to examine whether what we are doing is correct as the model of the brain, then what we are doing it have meaning because so when we want to do some tasks for animals, non-human animals, only they can do this, just a simple task. And to my belief, to my experiences, most of those tasks can be implemented by simple POMDP architecture, and some require some extension. But basically, what we are dealing with here is sufficient to characterize what animals are doing in the system neuroscience sense. So yeah, beyond that, we may be able to consider some more functioning neural networks through our strategy, but it is not soon. So we need many other validation before going to that step.

What does the reverse engineering, as you have it in the GitHub, or as you imagine this developing, what does it yield that a systems biology assessment wouldn't?

You mean, what is, sorry, can you rephrase that?

Yeah, what do we gain and learn from the reverse engineering that just a circuit level dissection doesn't provide?

Yeah, so circuit level dissection basically provide us static information, right? Information about architecture, neural population at that point. So we see a snapshot. So what we are interested in is more

prediction things. So through reverse engineering, we get a genetic model, which has, so it has learning ability, it itself is agent. So instead of making connectomics or making some the characteristic of static neural network, we can characterize the dynamics of neural network in terms of a genetic model. It provides us a prediction of long term future after some running. So which is completely different from what a conventional system neural science does.

Okay, another question, I put it in the chat. So how do we reconcile the traditionally, essentially error-free nature of Turing machines and digital computation with the more fundamentally probabilistic nature of neural and Bayesian systems? Like part of the reason why the symbolic architectures are useful is because they can be used deterministically. So are we seeking something that's deterministic or probabilistic? Turing them to be able to do that. So it is related to our demonstration using others and hand digit numbers. So hand digit are your noisy stochastic information. And once we, our network is going to extract the hidden features from such a stochastic information, then hidden layer is become becoming more deterministic in the sense that if we accumulate sufficient information, the POMDP model

becomes either zero or one state, right? So in that case, the state transition can be very deterministic. Of course, we, in general, we need to deal with the stochastic environment. But yeah, sometimes either likelihood matrix or transition matrix can be deterministic. And under that situation, our machine provides a very accurate prediction about states, current state or future states. And moreover, although a conventional naive Turing machine is very deterministic or discrete, but the financial Turing machine incorporates the current continuous dynamics and discrete Turing machine. So it is a Turing machine that is expressed by the differential equation. And we extend that idea to the energy minimization in the sense that differential Turing machine of us is derived as the gradient descent on some energy. So this dynamics is something between the discretized symbolic computation and continuous dynamics. And based on the condition, just like

inverse temperature parameter, we can shift that dynamics both to discretize symbolic computation and continuous dynamics.

That idea underlying what we are doing. Great. Okay. So in computer science theory, the halting problem or Turing's halting problem is

the description that or the problem that the description of a computer program and the input, you cannot determine whether the program will finish running or run forever. So if we can make this kind of software 2.0

neural network representation and or like this software question mark with the Bayesian interpretation as well, are they also bound by some of these constraints? Or do you think there could be some angles of approach where problems that are

not solvable in a procedural computational framework could be approached probabilistically and vice versa?

Yes, thank you for the question. Yeah, the simple answer is I don't know the exact answer, but so I tried to make my opinion.

So, yeah, first of all, although we consider some neural networks that perform Turing machine, so it is still in the framework of Turing machine or a chart Turing Tese in the sense that,

you know, what we are doing is just computation based on algorithms, right? Even though we are designing very realistic brain model comprising Hosenfield-Hackstray spiking neural network, so it isn't just an algorithm, so it is implemented in terms of Turing machine. So in that sense, nothing different. But, and it means that we do not solve the halting problem. So our network can conclude whether this program is finished or not. However,

we can guess whether this is a good strategy or a bad strategy. So one interesting difference from conventional Turing machine is that this, our neural network works in real time. So time is passing during computations, unlike Turing machine, a conventional Turing machine. So which means that if we select a bad strategy that does not stop during the meaningful time frame, then that animal, that synthetic animal die or eliminated through the natural selection stage. So survived species should be should have some meaningful function, meaningful algorithm that works in a realistic time frame.

So yeah, that works as some debug or selection of algorithm. So that is what I'm thinking.

Yeah, the connections with the neural Darwinism are very interesting. And also the open-endedness of successful life, it's like those are the programs that haven't halted. And you don't know whether the environment could change, the environment could change, and they could all of a sudden die. But what you know by

observing their existence is that they haven't halted yet. So it's like one sense they're able to carry out a computation in plausible time. And on the other hand, their persistence as a species, as a population, is still running. So how will you proceed from here? What directions do you plan to follow?

The plan involves some emergence of new algorithm. I mean, yeah, algorithm, so now we have mean to implement represents arbitrary algorithm within the neural network architecture, which is corresponding to extended POMDP. And what now we are interested in is whether the function that can, the function those network can do is only limited to imitation of external world, or it can do something different from the simple imitation of external world. So if the internal algorithm itself has some new meaning, then that would be more interesting.

What would that look like? Or how will you explore that?

Yeah, it's a difficult question.

And what can happen within a fixed topology of a Bayes graph or neural network or program information flow?

And then what is possible when the structure of the network changes? Because there's a space of semantics that can be explored with the quantitative variation, just through the weight updating.

And then there's another shell of possibility based upon topological change.

That's true. That's true. So for example, now we do not involve structural plasticity within a lifetime, but the model selection done by update of the gene involves the change of the size of network.

You know, by changing the gene, we can freely

change the number of neural involving network or whether it is a two layer or three layer, like that.

So anything can be characterized by the gene and its selection means the model selection from various model types.

So, I mean, yeah, in principle, we can consider many things, but question is whether it is plausible, I mean, yeah, intuitively it requires very high computational cost, you know, it is something like a random search in the huge binary space.

So as you may know, yeah, combinatorial optimization takes exponential time.

So question is whether animals do that or animals know better searching algorithm or animal avoid to do that.

And yeah, how another solution. So that's kind of things is very interesting to me.

Cool. Is there anything else you want to add?

Yeah. So although I'm not fully sure whether we can validate the Turing machine, relationship between Turing machine, Bayesian inference and neural network using real data, at least we can validate the double equivalence between neural network and Bayesian inference.

And we can extend some something to the evolutionary direction or Turing machine direction.

So, yeah, I think as the neuroscience, I think I'm a theoretical neuroscientist.

So as the neuroscience, we need to do both developing development of the theory and validation of theory.

So I think I'm a little bit of learning to do both of these.

So I'm very interested in both sides.

And if you are interested in this sort of research, I'm very happy to working with you.

Thank you.

Last question.

Do you think there's a fourth player at that intersection?

Yeah, that's a very interesting question. So one thing which I used to be interested in is consciousness things.

So yeah, many people nowadays are interested in consciousness and there are several theories of consciousness.

But as long as I know, those three pillars, neural network, Bayesian influence and the Turing machine algorithm computation, do not involve consciousness idea.

So I want to ask some expert that whether consciousness can be involved in those framework or it is different from those framework.

Maybe that can be a false pillar.

Awesome. Okay. Thank you very much for joining.

It's exciting to see how things have developed with the empirical work and this new theoretical layer as well since a few years ago.

So looking forward to where you take it from here.

Thank you very much.

Thank you. Bye.

Bye.